



Project Atomic-Link: Data Architecture

Bloc 2 Defense
Aurora Tech Computing Group



LOGISTICS



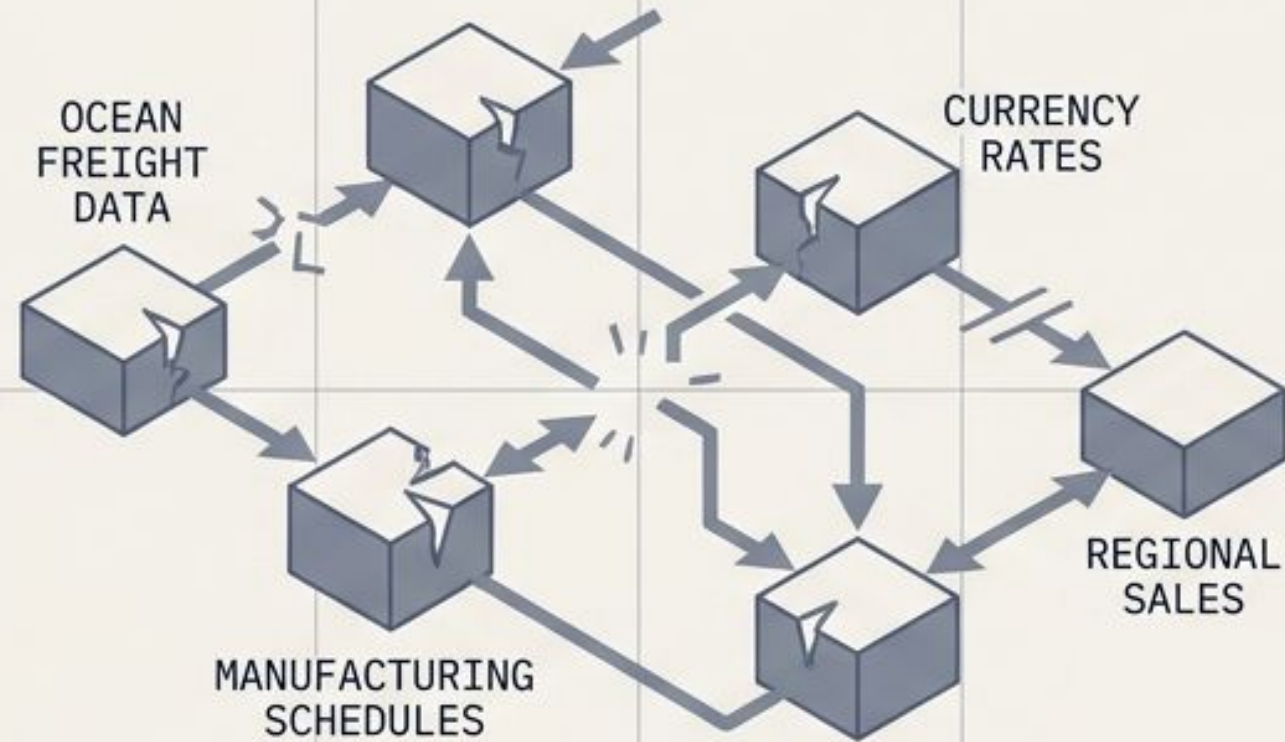
FINANCE



AI

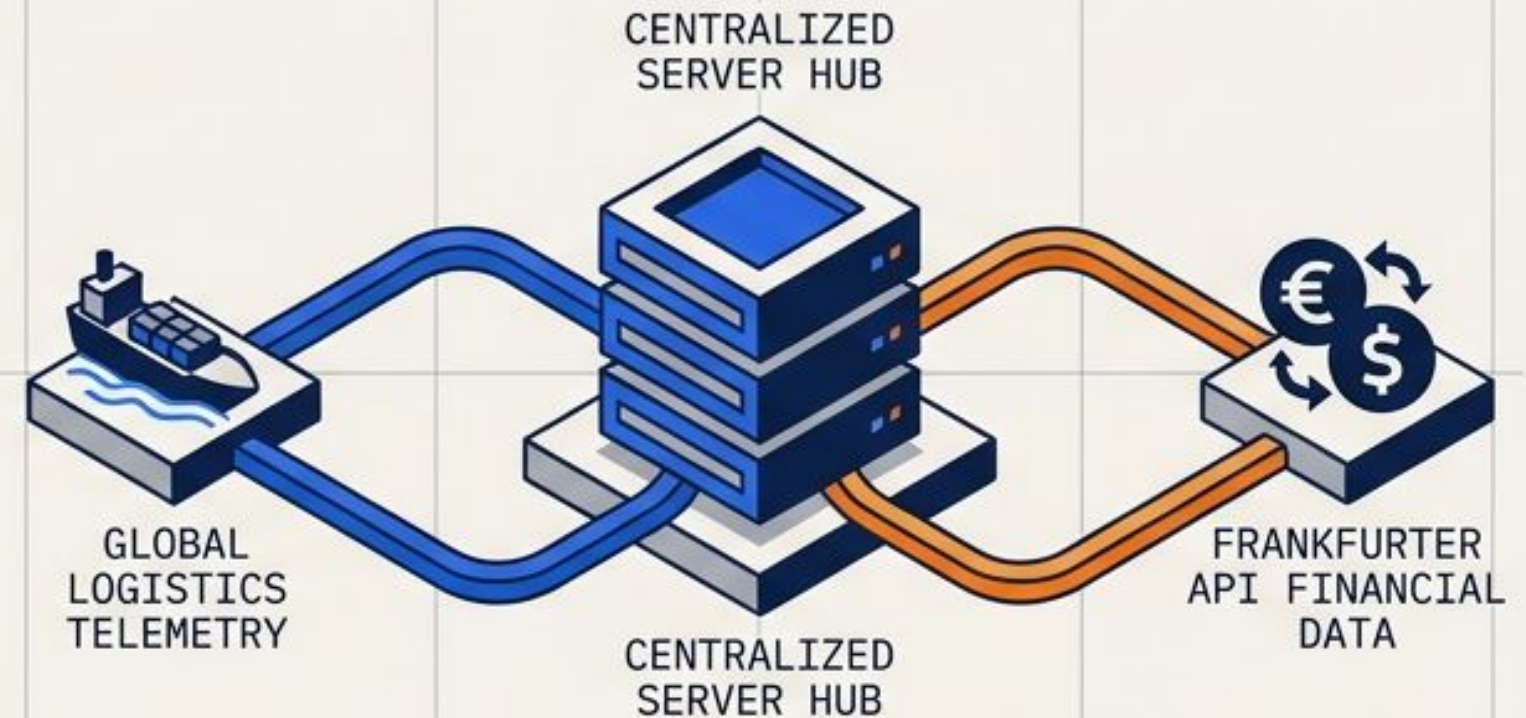
Solving margin compression with unified infrastructure

The Problem



- Shipping delays (Ocean-to-Air conversions)
- Volatile currency shifts (EUR/USD/TWD)
- Blind spots causing profit margin loss on Chromebooks and NVIDIA AI Workstations

The Solution



- **Real-time** Frankfurter API financial data fused with global logistics telemetry
- **Lightning-fast**, centralized server hub
- **Predicts costs** before they happen to save profit margins

The four pillars of the Atomic-Link system

Data Ingestion

Apache Airflow

Pulls live FX and freight data.



Data Storage

PostgreSQL

The heavy, stable core of the system



AI Engine

FastAPI

Real-time predictive margin modeling.



User Dashboard

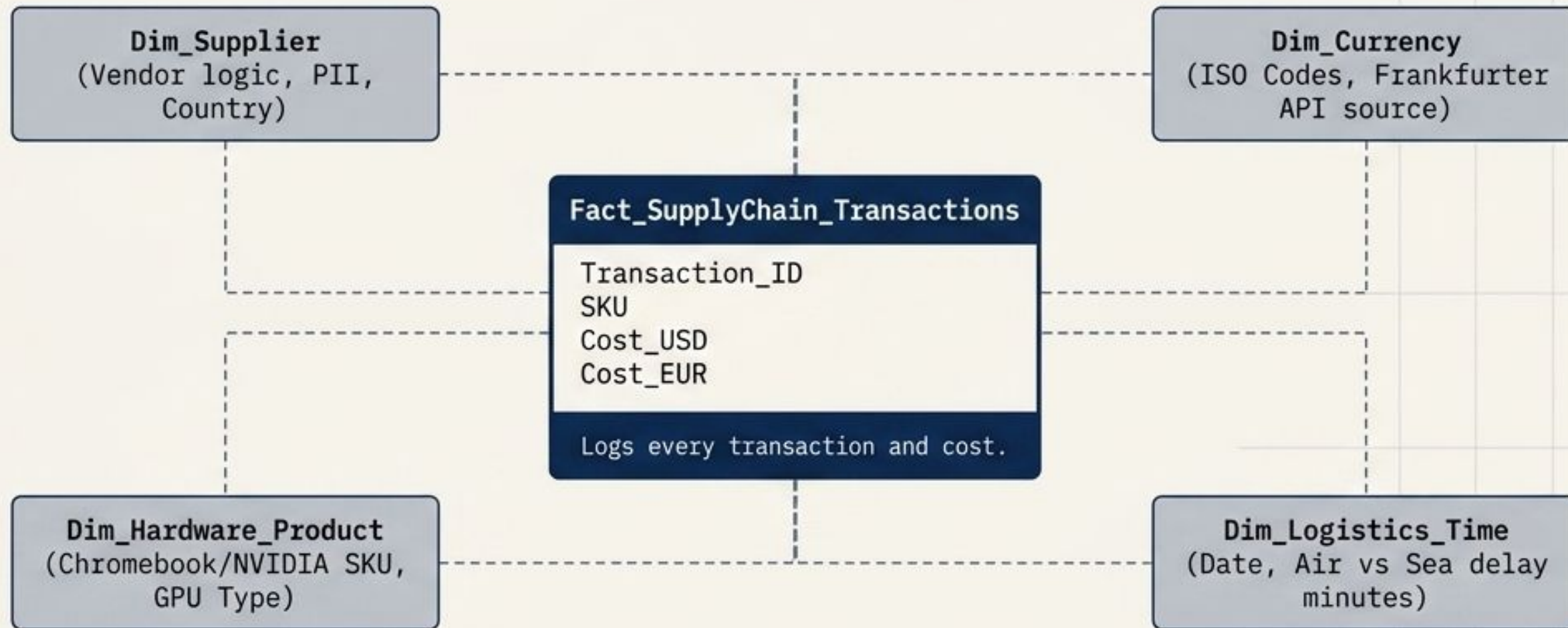
React

Global operations command center.



Core Insight: Every component is modular, containerized, and built to scale globally.

A clean star schema engineered for instant AI queries



Logical data isolation ensures fast reads for the Random Forest model while protecting sensitive supply chain pricing.

Automated global deployment from code to cloud



 Global Load Balancer

HEALTHY / 4 REGIONS

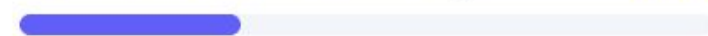


 Application Tier (EKS)

Core API CPU Load 65%



Worker Nodes Memory 32%



 Transit Pipeline Layer

Kafka Topics Throughput 8.2k msg/s



Redis Cache Hit Rate 94%

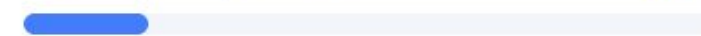


 Data Warehouse (PostgreSQL)

Storage Provision 1.2 TB / 5 TB



Live Query Load 18%



[Enter DW Explorer →](#)



SCHEMAS

PostgreSQL

dim_vendor

fact_margin_risk

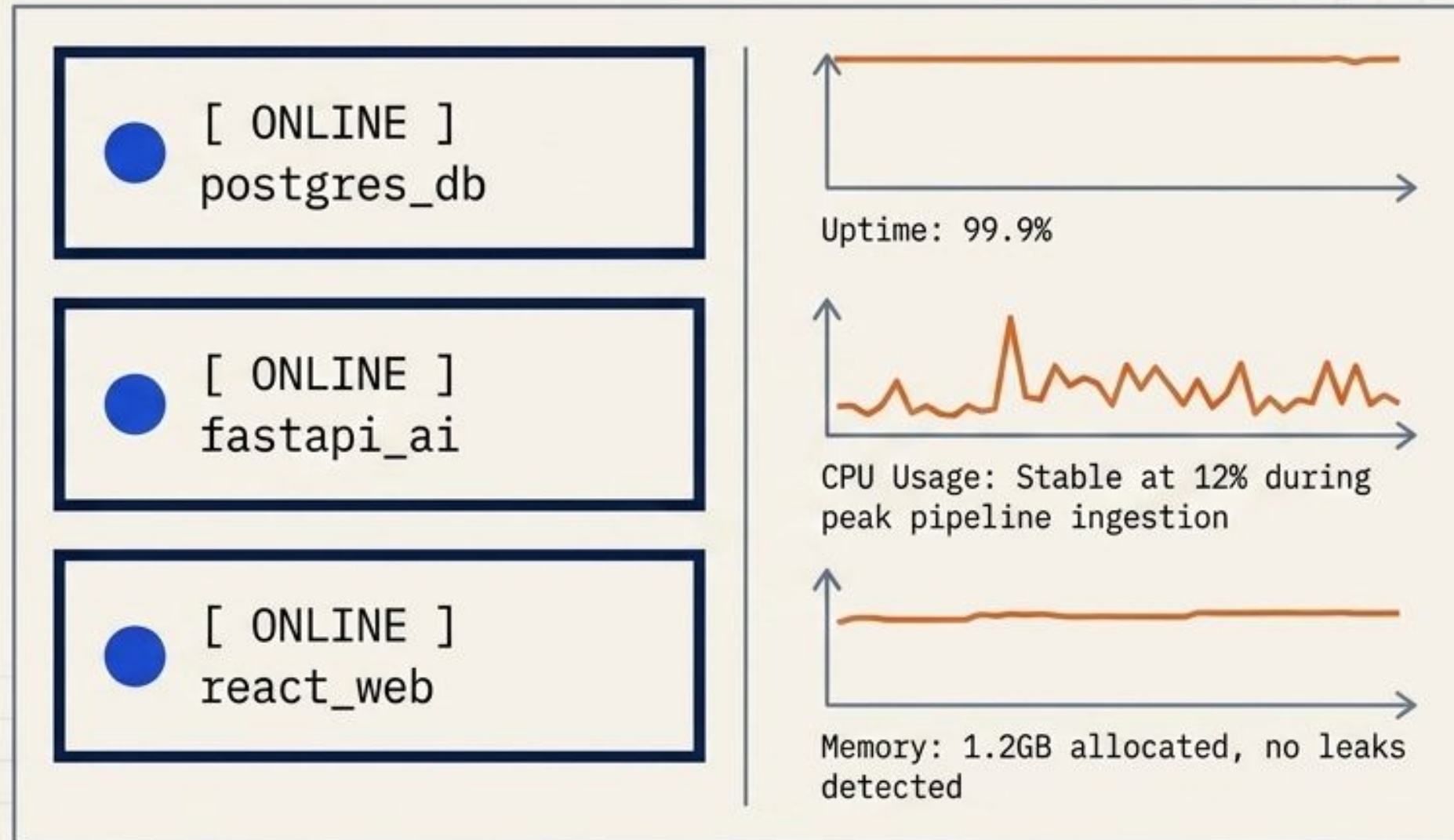
dim_date

```
SELECT * FROM dim_vendor
WHERE region = 'Taiwan';
```

Execute

ID	VENDOR	REGION	TYPE
VND-NV-01	NVIDIA Corporation	Taiwan	GPU
VND-TSMC-02	TSMC Foundry	Taiwan	Semiconductor
VND-AUO-03	AU Optronics	Taiwan	Display
VND-INT-04	Intel	USA	CPU
4 rows retrieved		Execution time: 1.2s	

Live observability ensures zero data loss



"We cannot fix what we cannot see. Instant alerts ensure Aurora Tech never loses a single supply chain data point."